

RAYMOND SHAW

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EDUCATION

The University of Texas at Austin, Austin, TX *Expected Graduation: May 2028*
Bachelor of Science, Computer Science | Minors: Economics, Japanese | Current GPA: 3.96
Relevant Coursework: Speech/Audio Processing, Operating Systems, Computer Architecture, Linear Algebra

SKILLS

Technical Skills: Python, Java, C, SQL, JavaScript, PyTorch, Scikit-learn, Hugging Face, React, Node.js, LLM, Computer Vision, Data Schema Design, Data Pipelining, Full-Stack Development, Chinese, Japanese

EXPERIENCE

AMRL Lab, Austin, TX | *Undergraduate Researcher* *Sep 2026 - Present*

- Researching Omniword model integration to assist in social navigation for autonomous robotic pathing.
- Training a Cosmos-3 model to output socially acceptable primitives for ComposableNav diffusion planning.
- Developing supervised fine-tuning and heuristic evaluation pipelines with ROS 2 for robotic simulation.

MetOX International, Houston, TX | *AI/ML and Data Analyst Intern* *Jun 2026 - Aug 2026*

- Developed ML models to predict manufacturing indicators and support automated machine adjustments.
- Built web applications to aggregate, visualize, and adjust machine data through interactive charts and graphs.
- Overhauled defect detection with YOLO and PatchCore CV, uncovering previously undetected product defects.
- Produced ETL pipelines to transfer high-volume PLC signal and MySQL database data for custom alert systems.

McCombs AI/ML Business Application Program, Online | *Program Participant* *Dec 2025 - Sep 2026*

- Built a Qwen-based medical assistant LLM chatbot with RAG procedures using PyTorch and Hugging Face.
- Developed and evaluated financial prediction and fraud detection models with Scikit-learn and data engineering.
- Trained a Keras CV model for safety hazard detection using normalized and preprocessed images.

MAG Neurospine Clinic, Houston, TX | *Web Developer Intern* *Jul 2025 - Aug 2025*

- Developed a database web app to manage and securely access patient data entries into Microsoft Dataverse.
- Automated the migration of legacy PDF and Excel data to Power Apps using SQL and Power Query.
- Prepared technical reports to streamline communication with stakeholders on planning and execution.

NASA High School Aerospace Scholars Program, Houston, TX | *Project Manager* *Sep 2023 - Jun 2024*

- Partnered with NASA engineers and led a 10-person research group to simulate a lunar landing mission.
- Integrated CAD modeling and programmed simulations for lunar base prototype designs

PROJECTS

Kanadle - Web Application | [Link](#) *Sep 2026 - Present*

- Developing a multi-layered full-stack application using JavaScript and React for interactive UI and state logic.
- Employs a Node.js backend integrating Jisho API requests to receive and structure dynamic game logic.

Japanese Word Bomb - Online Web-based Multiplayer Game | [Link](#) *Nov 2025 - Jan 2026*

- Built a real-time Japanese word query multiplayer game using Python WebSockets, FastAPI, and Pydantic.
- Designed custom network protocols, private lobby system, and synchronized game state for smooth gameplay.

ORGANIZATIONS

DiRP, Austin, TX | *LLM-Training Mentee* *Jan 2026 - Present*

- Applied transformer, RoPE, and tokenization techniques with PyTorch to develop and train a Nano-LLM Chatbot.
- Studied and reproduced modern model innovations such as MoE architecture and memory pipelining.

ECLAIR, Austin, TX | *Development Member* *Oct 2025 - Dec 2025*

- Contributed to Trace, a rotating platform that guides an object along a digitally drawn path.
- Built a custom Python drawing interface and mapped drawings and velocities through Bezier curves.

Elkins HS Math and Science Club, Missouri City, TX | *Vice President* *Sep 2023 - May 2025*

- Co-led and won with 4 state-level UIL/TMSCA competition math teams (first wins since 2004).
- Planned and taught twice weekly training sessions and built custom TI-Nspire calculator functions.

AWARDS

- 1st place team - UIL State Competition (Mathematics and Calculator) *May 2025*
- 1st place team - TMSCA State Competition (Mathematics and Calculator) *Mar 2025*
- 2nd place, 3rd place - University of Houston Math Contest *Feb 2023, Feb 2024*